

Another large strategic investment in 2018 in Extruded Solutions was breaking ground for the construction of a new high-performance extrusion press at the Cressona plant in Pennsylvania in the U.S. The new press will have state-of-the-art capabilities primarily targeted at meeting the needs of the growing automotive market in North America. The amount of aluminium content per vehicle in the U.S. automotive industry is projected to increase as OEMs are demanding stronger, light-weight applications that help

improve fuel economy and payload. The new press in Cressona will be able to produce new higher strength alloys for applications that need additional value-added fabrication and will provide Extruded Solutions in North America with a solid platform for future business development. The plant has sound financial performance, an established automotive culture, and vast technical resources to successfully deliver on this expansion project and capture new business opportunities in North America.

Our strategy is about “value over volume” driven by innovation, collaboration and simplification

<i>Better</i>	<i>Bigger</i>	<i>Greener</i>
Raise performance and improve customer offering	Expand the use of aluminium and strengthen Hydro's platform for growth	Lead the transition towards sustainable solutions
· Employer of choice with world-class safety, quality, people development and compliance · Higher value creation through innovation and collaboration · Strong profitability and value through commercial, service, and operational excellence	· Develop leading positions in select profitable markets · Expand value-added capabilities to strengthen our offerings to customers · Drive material substitution to aluminium in new and existing applications	· Promote recyclability and aluminium applications for a low carbon, circular economy · Increase recycling footprint · Reduce emissions and waste throughout our value chain · Support communities through CSR initiatives

Operations

Extruded Solutions is the leading global supplier of extrusion-based aluminium solutions with a market share of 20 percent in Europe and 23 percent in North America. The business area also has a solid foothold in emerging markets with extrusion capacity in South America and Asia. Extruded Solutions has four business units: Extrusion Europe, Extrusion North America, Precision Tubing and Building Systems. The business units are responsible for their respective value chains, from casthouses, aluminium extrusion and value-adding operations to commercial activities such as product development and sales.

Extruded Solutions has an extensive network of production plants that ensures global reach combined with local presence. During 2018, Extruded Solutions focused on simplification, and adding more value and locating resources closer to customers.

Cost and revenue drivers

Extruded Solutions is a margin business and the LME aluminium cost element is passed on to the customer. Contracts are typically short to medium term. Extruded Solutions will continue to shift its portfolio toward higher-margin products.

Competitive strengths

- Strong technology competence with dedicated R&D centers and more than 1,000 engineers close to customers
- Global reach and local presence, increasing flexibility and reliability for customers
- Speed in delivery and proximity to customers
- Value chain breadth and depth, from extrusion to surface finishing and fabrication, welding and assembly
- Strong and agile product development capabilities
- Technical leadership in precision tubing and extrusion

Technology and innovation

Extruded Solutions' innovation model promotes fast decision paths from idea to product. The business area has four R&D centers, located in Finspång (Sweden), Troy (USA), Karmøy (Norway) and Toulouse (France). These are complemented by application centers covering nearly 40 locations across the four business units and by corrosion labs in China, the U.S. and Norway.



All efforts are shared across business units to achieve optimal knowledge transfer and to enable faster, better impact in other markets where similar opportunities arise.

To deliver better products to the customers faster, and to address specific challenges, Extruded Solutions collaborates with leading universities such as the Norwegian University of Science & Technology, Michigan Technological University, Massachusetts Institute of Technology and the University of Oxford, as well as with specialized companies.

Building Systems has opened physical showrooms in selected cities in Europe to promote their latest product developments. Product innovation is vital to success in that market, mainly in Europe, where end-users are becoming advanced.

In Precision Tubing, the innovation is focused on developing new aluminium applications for the automotive industry, such as fuel and brake lines. Aluminium substitutes copper and steel, resulting in comparable performance, but lighter